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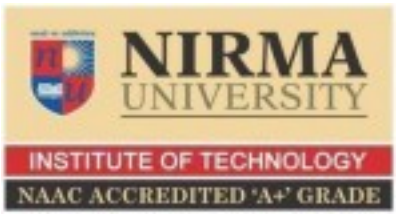
Software Project Management - 4CS504ME25 B.

Tech. Semester IV

Prepared By :

Gandhi Urva Yogeshkumar (23BCE078)

Gajnotar Rakshit (23BCE077)



Institute of Technology,
Nirma University
Ahmedabad - 382481

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Aim

To implement project monitoring and tracking tools including a Project Status Report, Earned Value Analysis (EVA), Milestone Tracking Chart and Defect Tracking Dashboard for the Hospital Management System (HMS) at a status date of Week 12.

Objective

- Compare actual project performance against the 16-week baseline schedule (T5), effort estimates (T7) and defect baselines (T6).
- Establish a Red Amber Green (RAG) status threshold system for project health indicators.
- Perform Earned Value Analysis calculating PV, EV, AC, Schedule Variance, Cost Variance, SPI and CPI.
- Track deliverable milestones and measure schedule slip across Milestones M1 through M6.
- Build a Defect Tracking Dashboard analyzing closure rates, defect density, backlog trends and critical open bugs.
- Identify project deviations and formulate a fast-track recovery plan to protect the final Week 16 delivery date.
- Forecast final effort and cost at completion using EAC and ETC formulas.

1. Introduction

Planning a project is only the first step of project management. Once development starts, the manager must continuously check whether work is following the plan and take quick corrective action whenever things slip. This process is called project monitoring and tracking.

In this tutorial we monitor the Hospital Management System using our earlier baselines: the 16-week CPM schedule and milestones from T5, the 8.0 person-month effort estimate from T7, the defect and quality planning figures from T6 and the risk register from T9.

We review the project at a status date set at the end of Week 12 (conclusion of Sprint 5). At this point, five of the six core modules have been coded and integrated. We evaluate status using three monitoring tools: a Project Status Report with Earned Value Analysis, a Milestone Tracking Chart and a Defect Tracking Dashboard.

2. Monitoring and Tracking Framework

Our monitoring framework tracks five dimensions: Schedule, Effort and Cost, Milestone Completion, Defect Quality and Risks. Each metric is evaluated against fixed RAG thresholds agreed in advance:

Status Indicator	Condition / Threshold	Action Required
Green (On Track)	Deviation within +/- 5% of baseline plan (SPI >= 0.95, CPI >= 0.95)	Proceed according to regular sprint plan.
Amber (Caution)	Deviation between 5% and 10% behind plan (SPI 0.90 to 0.94, CPI 0.90 to 0.94)	Apply sprint recovery actions and use buffer.
Red (Critical)	Deviation > 10% behind plan (SPI < 0.90, CPI < 0.90, critical milestone slip)	Escalate to mentor and initiate replanning.

3. Project Status Report (Status Date: End of Week 12)

At the end of Week 12, Sprint 5 (Billing and Invoice module) concluded. While Patient, Doctor, Appointment and Pharmacy modules are operational, testing the billing calculation engine and edge cases with prescription discounts caused minor delay.

3.1 Earned Value Analysis (EVA)

Earned Value Analysis connects schedule progress and effort in one integrated view:

Metric Name	Abbreviation	Formula	Calculated Value	RAG Status
Budget at Completion	BAC	Total Planned Project Budget	8.00 Person-Months	Baseline
Planned Value	PV	Planned % * BAC (12 / 16 * 8.00)	6.00 Person-Months	Baseline
Earned Value	EV	Completed Work % * BAC	5.60 Person-Months	Amber
Actual Cost	AC	Actual Effort Expended to Date	6.20 Person-Months	Amber
Schedule Variance	SV	EV - PV (5.60 - 6.00)	-0.40 Person-Months	Amber
Cost Variance	CV	EV - AC (5.60 - 6.20)	-0.60 Person-Months	Amber
Schedule Performance Index	SPI	EV / PV (5.60 / 6.00)	0.933 (~0.93)	Amber
Cost Performance Index	CPI	EV / AC (5.60 / 6.20)	0.903 (~0.90)	Amber
Estimate at Completion	EAC	BAC / CPI (8.00 / 0.903)	8.86 Person-Months	Amber
Variance at Completion	VAC	BAC - EAC (8.00 - 8.86)	-0.86 Person-Months	Amber (10.7%)
Estimate to Complete	ETC	EAC - AC (8.86 - 6.20)	2.66 Person-Months	Within Reserve

Analysis of EVA Results

- Schedule Performance: An SPI of 0.93 indicates the project is progressing at 93% of planned speed, representing roughly 0.40 person-months (about half a week) of schedule slip.
- Effort Efficiency: A CPI of 0.90 shows that for every person-month spent, 0.90 PM of planned work was earned. Additional time was spent resolving database locking in billing transactions.
- Forecast at Completion: EAC projects total effort at 8.86 PM. The forecast overrun of 0.86 PM (10.7%) is fully absorbed by the 10% to 15% contingency reserve established in T7, leaving the project financially viable.

4. Milestone Tracking Chart and Analysis

Tracking milestones against the baseline CPM schedule shows whether delivery delays propagate into downstream sprints:

Milestone	Baseline Week	Actual / Forecast	Slip (Weeks)	Status
M1: Inception and Architecture Frozen	Week 2	Week 2 (Actual)	0.0	Achieved on time
M2: Patient and Doctor Operational	Week 6	Week 6 (Actual)	0.0	Achieved on time

Milestone	Baseline Week	Actual / Forecast	Slip (Weeks)	Status
M3: Appointment Booking Live	Week 8	Week 8 (Actual)	0.0	Achieved on time
M4: Pharmacy and Billing Verified	Week 12	Week 12.5 (Actual)	+0.5 week	Achieved 0.5 wk late
M5: Dashboard and All Modules Coded	Week 14	Week 14.5 (Forecast)	+0.5 week	At risk (Recoverable)
M6: Final Release and Submission	Week 16	Week 16.0 (Forecast)	0.0	On track with recovery

- Milestone Metrics: 4 of 4 milestones due by Week 12 have been achieved (100% completion rate), with 3 of 4 on time (75%).
- Critical Slip: The 0.5-week slip at M4 was caused by complex tax and discount logic in billing invoices. Because Billing sits on the critical path (T5), this delay pushed the forecast start of M5 back by half a week.
- Recovery Path: Because Sprint 7 contains integration and documentation buffers, fast-tracking integration during Sprint 6 allows M6 to finish exactly at Week 16.0 as planned.

5. Defect Tracking Dashboard

The Defect Tracking Dashboard summarizes testing progress, closure velocity and remaining open issues:

Indicator	Value at Week 12	Formula / Context
Total Defects Logged	32 Defects	Recorded since testing began in Sprint 1 (Week 3).
Total Defects Resolved	26 Defects	Verified and closed in sprint bug fixes.
Currently Open Defects	6 Defects	Active defects: 1 Critical, 2 High, 2 Medium and 1 Low.
Defect Closure Rate	81.25%	Closed / Found = 26 / 32 * 100.
Net Backlog Change This Week	+1 Defect	5 found and 4 closed during Sprint 5 review.
Defect Density So Far	12.8 Defects / KLOC	32 defects in approx 2.5 KLOC written to date.
Share of Expected Defects	66.7%	32 of 48 total errors anticipated before release (T6).

Important Open Defects (Critical and High)

The table below details all open defects requiring resolution prior to Sprint 6 admin module completion:

ID	Module	Severity	Description	Age (Days)	Owner	Target
D-019	Billing	**Critical**	Invoice total does not recalculate when a medicine row is deleted	3	Urva	Week 13 (24h)
D-022	Appointment	High	Allows booking appointment on a doctor's registered weekly day-off	4	Rakshit	Week 13
D-024	Pharmacy	High	Rapid concurrent checkout can	5	Urva	Week 13

ID	Module	Severity	Description	Age (Days)	Owner	Target
			decrement stock quantity below zero			
D-027	Patient	Medium	Phone number validation rejects valid 10-digit formats with spaces	2	Rakshit	Week 14
D-029	Doctor	Medium	Department filter dropdown does not reset after searching doctor name	3	Rakshit	Week 14
D-031	Billing	Low	Invoice print preview clips right margin border on smaller displays	6	Rakshit	Week 14

- Defect Density: 12.8 defects per KLOC written so far is consistent with our planned 14.55 errors per KLOC from T6.
- Backlog Trend: The open backlog of 6 defects is manageable. However, the 1 Critical (D-019) and 2 High defects (D-022 and D-024) must be closed within 48 hours to avoid blocking admin reporting.

6. Deviations Identified and Corrective Actions

Monitoring is useful only when it produces concrete corrective actions. The table below lists all deviations detected at Week 12 along with assigned owners and deadlines:

#	Deviation Identified	Root Cause	Project Effect	Corrective Action Decided	Owner / Due
1	Billing testing slipped 0.5 week (SPI 0.93)	Complex pharmacy tax and discount rules in invoice calculation	Pushed Admin Dashboard start back by half a week	Fast-track: start admin frontend UI in parallel while finishing billing testing; freeze requirements	Urva / Wk 13
2	Effort over plan by 0.60 PM (CPI 0.90)	Extra database locking and transaction debugging in billing	EAC rises to 8.86 PM, utilizing 10.7% reserve	Log effort daily; adopt strict unit test checklists before merging API changes to reduce rework	Both / Weekly
3	Critical defect D-019 open for 3 days	Event listener missed deletion event in React table state	Doctors and patients see incorrect billed amount	Fix invoice state recalculation within 24 hours; add automated Cypress test for deletion	Urva / Wk 13
4	Concurrent appointment race risk (D-022 and R7)	Missing row lock on doctor slot during booking submit	Risk of double-booking same doctor time slot	Implement atomic PostgreSQL transaction with SELECT FOR UPDATE on doctor appointment slots	Urva / Wk 13
5	Scope creep: requests for inpatient bed tracker	Clinical staff requested extra bed management screen	Threatens to derail Sprint 6 admin delivery	Enforce scope freeze; document bed tracker in future roadmap backlog for Version 2.0	Rakshit / Wk 13
6	DRE target of 90% requires protection	32 of 48 expected errors found so far	Risk of defect leakage into hospital pilot trial	Dedicate full second week of Sprint 7 to joint regression testing across all 6 modules	Both / Wk 15

7. Forecast and Recovery Outlook

Based on our Week 12 tracking data, we compare the project outlook with and without the fast-track recovery plan:

Scenario	Final Completion Date	Forecast Effort	Forecast Cost	Assessment
Without Recovery Plan	Week 16.5 (+0.5 week late)	8.86 PM	₹1,99,350	Late delivery; exceeds semester deadline.
With Fast-Track Recovery	Week 16.0 (On Schedule)	8.86 PM	₹1,99,350	Delivers on time; fits inside 10% reserve.

The recovery plan succeeds because the final sprint (Sprint 7) was designed with testing and documentation buffers. By resolving critical open defects within 48 hours and running integration checks concurrently with admin screens in Sprint 6, the project will finish on the baseline date of Week 16.0.

8. Conclusion

Monitoring the Hospital Management System at Week 12 with a status report, milestone tracking and defect dashboard provides clear visibility into project health. The Earned Value indicators (SPI 0.93 and CPI 0.90) show minor schedule slip and effort variance, caused by billing calculation complexities. Because these deviations were detected at Week 12, the team has time to execute the recovery plan, clear the 6 open defects and deliver the final release on schedule at Week 16.0 within our planned contingency reserve.